CODEGNAN — APTITUDE MOCK INTERVIEW

A. Percentages, Profit & Loss

1. A number is increased by 25% and then decreased by 20%. Find the net percentage change.
2. A student scores 30% below the pass mark. After gaining 40 additional marks, the student scores 10% above the pass mark. If the pass mark is 40% of the maximum marks, find the maximum marks.
3. An article is sold at a 15% loss. If it were sold for ₹255 more, the profit would be 10%. Find the cost price.
4. A dishonest dealer uses a 900 g weight instead of 1 kg and charges the cost price of 1 kg. Find the profit percentage.

B. Ratio, Average & Ages

5. The incomes of A and B are in the ratio 5:7, and their expenditures are in the ratio 2:3. If both save ₹4,000, find their incomes.
6. The average age of 30 students is 20 years. When a teacher joins, the average becomes 21 years. Find the teacher's age.
7. The present ages of a father and son total 60 years. Six years ago, the father was five times as old as the son. Find their present ages.

C. Time & Work

8. A can complete a job in 12 days and B in 18 days. They work together for three days, after which C joins. C alone can complete the job in 24 days. How many additional days are needed to finish the remaining work?
9. Twelve workers can complete a project in 20 days. After five days, four workers leave. How many additional days will the remaining workers need?

D. Time, Speed & Distance

10. A train 180 m long crosses a pole in 9 seconds and a platform in 21 seconds. Find the platform's length.
11. A person walks to work at 5 km/h and arrives 12 minutes late. At 6 km/h, the person arrives 10 minutes early. Find the distance.
12. Two trains of lengths 150 m and 200 m travel in opposite directions at 54 km/h and 72 km/h. Find the time taken to cross each other.

E. Number System & Algebra

13. Find the remainder when 7^{103} is divided by 13.
14. Find the number of trailing zeros in 125!.
15. If $x + 1/x = 5$, find $x^2 + 1/x^2$ without solving for x.
16. If $a + b = 12$ and $ab = 32$, find $a^3 + b^3$.

F. Probability & Combinations

17. Two fair dice are rolled. Find the probability that their sum is 8.

18. A bag contains 5 red, 4 blue, and 3 green balls. Two balls are drawn without replacement. Find the probability that both are the same colour.
19. How many distinct arrangements can be made using all the letters of MISSISSIPPI?
20. In how many ways can a committee of four be selected from six men and five women if it must contain at least two women?

G. Mixtures & Interest

21. A 50-litre solution contains 30% acid. How much pure acid must be added to make the solution 50% acid?
22. A sum of ₹20,000 is invested at 10% per annum for two years. Find the difference between compound interest and simple interest, assuming annual compounding.

H. Logical Reasoning

23. Find the next number: 3, 8, 18, 38, 78, ?
24. Five people T, R, P, Q, and S sit in a row in that order. Who sits in the middle?
25. All developers are logical thinkers. Some logical thinkers are analysts. Can we conclude that some developers are analysts? Explain.
26. A person walks 10 m north, 6 m east, 10 m south, and 2 m west. Find the final distance and direction from the starting point.

I. Data Interpretation

Use the following data:

Department	Trained	Placed
Python	120	84
Java	100	75
Data Analytics	80	60
Full Stack	150	105

27. Which department or departments have the highest placement percentage?
28. Find the overall placement percentage across all departments.

J. Clocks & Critical Thinking

29. Find the smaller angle between the clock hands at 4:40.
30. Four people take 1, 2, 7, and 10 minutes to cross a bridge. They share one torch, and at most two people can cross at once. Find the minimum time required for all four to cross.